

Helix OTA — Per-Feature Coverage Matrix (§11.4.25 Coverage Ledger)

Revision: 1 **Last modified:** 2026-06-23T17:30:00Z **Scope:** the §11.4.25 Full-Automation-Coverage LEDGER for the helix_ota control plane + Android A/B stack — a per-FEATURE × per-test-type status matrix that consolidates the 2026-06-23 test-coverage + anti-bluff build-out program into the feature dimension the by-test-type docs do not surface. **Companion docs (referenced, NOT duplicated):** - By-test-type live ledger (§11.4.45) — MASTER_TEST_COVERAGE_PLAN_20260623.md - By-test-type audit-grade snapshot — PHASE1_COVERAGE_REPORT_20260623.md - Underlying evidence corpus — docs/qa/20260623-*/ (~28 run dirs)

Authority: the route/feature set is enumerated from server/internal/api/server.go (the live router) + the per-handler/per-test files in server/internal/api/; the A/B + agent features from the Android bricks under submodules/. **HEAD at authorship:** 1f6ec790. **Honesty (§11.4.6):** every non-empty cell cites the captured-evidence dir that backs it. GAP / PENDING / REFERENCE / N-A are stated as facts, never rounded up to green. Read this matrix together with §3 (gaps & honest findings) — the green cells are only real because the gaps are honestly marked.

1. How to read the matrix

- **unit** — Go unit / table tests with measured `go test -cover` (or JVM JaCoCo for Android features).
 - **integration (pgx)** — exercised against real Postgres booted on-demand (containers brick), `-tags integration`.
 - **e2e / live** — black-box against the real running system (real HTTP + real DB via F-CLUSTER), full user-equivalent flow.
 - **security** — trust-boundary / authz / JWT-tamper / fuzz / saturation as applicable to the feature.
 - **stress / chaos** — §11.4.85 sustained-load + failure-injection where the feature is on the hot path.
 - **benchmark** — benchstat-guarded micro-benchmark where the feature has a perf-critical Go function.
 - **Challenge** — registered in the helix_ota shell-dispatch Challenge bank (Phase 2).
 - Cell values: PASS / measured % / N-A (test type does not apply) / GAP (should exist, does not yet) / PENDING (blocked on external dep) / REFERENCE (proven on a reference target, not the production target).
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2. Per-feature × per-test-type matrix

#	Feature (route / surface)	unit	integration (pgx)	e2e / live
1	Health / readiness (/healthz, /readyz)	PASS 100%	PASS (booted)	PASS (/readyz→200
2	Auth / JWT (/auth/login, /auth/refresh)	PASS (api 90.5%)	PASS	PASS (login—used downstro
3	Device registration (/devices/register, list, by-hardware, :id/status)	PASS (device 92.6%)	PASS (store 85.5%)	PASS (device persisted, end run delete rea
4	Artifact upload + signature verify (/artifacts/upload, :id)	PASS	PASS	PASS (signed upload in 15/1 pipeline)
5	Delta generation (/deltas POST/GET)	PASS	PASS	PASS (delta p pipeline)
6	Release management (/releases POST/GET/:id)	PASS	PASS	PASS (release in 15/15)
7	Deployment (/deployments POST/GET/:id)	PASS	PASS	PASS (deploy in 15/15)
8	Rollout — staged + halt/evaluate (/rollout, /rollout/evaluate)	PASS (rollout 100% unit)	PASS (rollout 83.1%)	PASS (rollout in 15/15)

#	Feature (route / surface)	unit	integration (pgx)	e2e / live
9	Recall / rollback (/recall, /rollbacks)	PASS (handlers_recall_test)	PASS	partial (coverage handler+integration not in the 15/1 happy-path e2e)
10	Client update poll + anti-downgrade (/client/update)	PASS (handlers_client + antidowngrade test)	PASS	PASS (device receives-signals update in 15/1)
11	Telemetry ingest + views (/client/telemetry, /devices/:id/telemetry, /telemetry/overview)	PASS (schema 98.9%)	PASS	PASS (audit/telemetry rows persisted)
12	Groups / membership (/groups CRUD, /members)	PASS (handlers_group_test)	PASS	PASS (device_group end-of-run-de real)
13	Audit log (/audit)	PASS (handlers_audit_test)	PASS	PASS (audit_logs=1 rows persisted)
14	Projects (/projects CRUD)	PASS (coverage_project_deployment_test)	PASS	partial (unit+integration covered; not in operational 39 e2e flow)
15	Rate limiting / DDoS resilience (maxInflightMiddleware)	PASS (rate_limit_test)	N-A	N-A
16	Compression / middleware (gzip negotiation)	PASS (middleware_compression_test)	N-A	PASS (live H path)
17	Android agent (JVM) — ota-android-agent:core (poll + json)	PASS (LINE 100% / BRANCH 100%, 47 tests)	N-A	N-A
18	update_engine bridge (JVM) — ota-update-engine-bridge	PASS (LINE 100%, 113/113, 27 tests)	N-A	N-A

#	Feature (route / surface)	unit	integration (pgx)	e2e / live
19	A/B on-device flow (update_engine + AVB/dm-verity + auto-rollback, slot flip)	N-A (JVM unit ≠ on-device)	N-A	REFERENCE flip + rollback on Cuttlefish Tier-2)
20	cmd/* + tools/* binaries (ota-server, applyport, ota-device-emu, loadtest)	partial (applyport 24.4%, loadtest 63.3%, ota-device-emu 0.0%)	N-A	smoke (each r boots, vet+go clean)

Cross-cutting anti-bluff infrastructure (not a product feature, but the proof the matrix above is real — see the companion reports): F-ANTIBLUFF-LIB 8/8 mutation-proven (antibluff-lib); all 14 propagation gates + functional gates §1.1-paired (metagates, propagation-metagates); §11.4.165 independent verifier run on the batch and caught a tautological restore-integrity bluff (indep-verif-gate).

3. Gaps & honest findings (§11.4.6 — the anti-bluff proof)

These are the cells the matrix marks below-green, stated as facts, not papered over. They mirror the Phase-1 report §3; restated here against the per-feature dimension.

1. **Rate-limiter (feature 15) ships OFF by default.** The 429 shed/recover control is PROVEN to work when HELIX_MAX_INFLIGHT is set, but the shipped server/deploy/system.compose.yml leaves it **unset**, so maxInflightMiddleware is a no-op passthrough. The default stack survives a bounded 360-flood via host scheduling + bounded work, NOT via shedding. Recorded as a hardening recommendation (operator decision, §11.4.122) — never silently flipped. *Evidence:* docs/qa/20260623-saturation-live/case_b_SUMMARY.txt.
2. **A/B on-device flow (feature 19) is GAP / REFERENCE — no production hardware.** The JVM bricks (features 17–18) are LINE/BRANCH 100%, but the on-device update_engine + AVB/dm-verity + auto-rollback slot-flip is proven only on a **Cuttlefish cvd Tier-2 reference**, not on the RK3588 / Orange Pi 5 Max target. The cuttlefish-launch-verified run is itself **PARTIAL** — the privileged guest boot is operator-gated under rootless (expected VIRTUAL_DEVICE_BOOT_FAILED run_cvd returned 10); asset-feed + launch_cvd discovery + cvd config-assembly are proven, the privileged boot is

not. This is the single largest honest gap. *Evidence:* docs/qa/20260623-cuttlefish-tier2-ab/REPORT.md, docs/qa/20260623-cuttlefish-launch-verified/REPORT.md.

3. **Recall/rollback (feature 9) is PARTIAL at the e2e layer.** Covered by unit (handlers_recall_test) + integration (real pgx), but the 15/15 signed e2e pipeline exercises the happy-path deploy→rollout→poll flow, not a recall trigger end-to-end against the live system. No dedicated recall Challenge yet. *Next step:* a recall-halts-rollout live e2e + Challenge with paired mutation.
4. **Projects (feature 14) is PARTIAL at the e2e layer.** Unit (coverage_project_deployment_test) + integration covered; not part of the operational 39/39 black-box e2e flow, and no Project Challenge registered.
5. **ota-device-emu (feature 20) reports 0.0% unit coverage.** Recorded honestly: it is a dev/emulation tool — build-tested, boots, vet+gofmt clean (smoke-covered), but carries no unit tests. Not claimed as unit-covered. *Evidence:* docs/qa/20260623-cmd-smoke/evidence.txt.
6. **Challenge column is shell-dispatch only; several features show GAP.** Telemetry- schema (12/12), rollout-staged-halt (47/47), trust-boundary, load, chaos, and signed-pipeline are registered; delta, recall, groups, audit, projects have **no dedicated OTA Challenge** yet. The Go cmd/helixqa orchestrator is **OPERATOR-GATED** (7 of 8 own-org replace submodules missing + §11.4.28(C) layout mismatch) — bank format matches, needs a ~1-file DeviceExec adapter. *Evidence:* docs/qa/20260623-phase2-challenges/, docs/qa/20260623-helixqa-orchestrator/WIRING_PLAN.md.
7. **Functional media/vision gates (§11.4.163/.165) are LATENT.** helix_ota ships no UI surface; the media-validation pipeline binds the moment a UI/recorded-artifact surface ships (§11.4.96 pattern). Recorded as latent, not as a failed gate.
8. **Structurally-unreachable telemetry branch left honestly uncovered.** ota-telemetry- schema is **98.9%** — the 1.1% residual is a decode branch not reachable through the public path; left uncovered rather than forced to a synthetic PASS. *Evidence:* docs/qa/20260623-ota-coverage/ota-telemetry-schema_after.txt.

4. One-line verdict per feature

#	Feature	Verdict
1	Health / readiness	Fully covered across applicable types; live-boot + load proven.
2	Auth / JWT	Fully covered; trust-boundary + tamper rejection proven.
3	Device registration	Fully covered; race-idempotency proven (exactly-one-device).

#	Feature	Verdict
4	Artifact upload + signature verify	Fully covered; trust boundary 4/4 + fuzz 0-crash + bad-sig Challenge.
5	Delta generation	Covered incl. benchmark guard (196ns/0-allocs); no dedicated Challenge.
6	Release management	Fully covered through the signed pipeline.
7	Deployment	Fully covered through the signed pipeline.
8	Rollout (staged + halt)	Fully covered; 47/47 staged-halt Challenge vs live.
9	Recall / rollback	PARTIAL — unit+integration solid; live e2e + Challenge are the gap.
10	Client update poll + anti-downgrade	Fully covered; device receives signed update in 15/15.
11	Telemetry	Fully covered; parser fuzzed + 12/12 schema Challenge vs live.
12	Groups / membership	Covered (unit+integration+e2e persistence); no Challenge yet.
13	Audit log	Covered; 14 rows persisted proof; no Challenge yet.
14	Projects	PARTIAL — unit+integration solid; e2e + Challenge are the gap.
15	Rate limiting / DDoS	Capability covered + proven, but ships OFF by default (hardening rec).
16	Compression / middleware	Covered under live HTTP + load.
17	Android agent (JVM)	Fully covered — LINE 100% / BRANCH 100%.
18	update_engine bridge (JVM)	Fully covered — LINE 100%.
19	A/B on-device flow	

#	Feature	Verdict
		GAP / REFERENCE — Cuttlefish Tier-2 only; no production hardware.
20	cmd/* + tools/* binaries	PARTIAL — smoke- covered; ota-device-emu 0% unit.

Tally: 20 enumerated features. - **Fully covered (across all applicable test types):** 14 — features 1–8, 10–13, 16, 17, 18. - **Covered-with-caveat:** 1 — feature 15 (capability proven but ships OFF; hardening rec). - **PARTIAL:** 3 — features 9 (recall e2e/Challenge), 14 (projects e2e/Challenge), 20 (emu 0% unit). - **GAP / REFERENCE:** 1 — feature 19 (A/B on-device; no production hardware).

The largest honest gap is the on-device A/B flow (feature 19), blocked on real RK3588 / Orange Pi 5 Max hardware; the JVM layer beneath it is at 100%. The remaining gaps are e2e/Challenge breadth for recall + projects, the emu unit-coverage hole, and the operator-gated rate-limiter default + HelixQA Go orchestrator.

This matrix is a faithful per-feature consolidation of the committed docs/qa/20260623- evidence corpus and the two companion reports. It asserts no result not backed by a cited captured-evidence dir. HTML + PDF siblings (§11.4.65) are generated by the conductor, not in this doc.*